Cyclohexane Reactor Ver 1 · operator training simulator

Equipment datasheets

Document no. CHX-V1-ME-DS-001 · **Discipline** Mechanical engineering

Client Department of Chemical Engineering, Faculty of Engineering, Chulalongkorn University (teaching)

• 15 datasheets, one per item, each numbered CHX-V1-ME-DS-<tag> (the number in the page header): R-101, R-102, R-103, E-101, E-103, E-106, D-101, D-102, V-103, V-104, F-102, P-100, P-101, P-102, K-101.
• Process data from the heat and material balance CHX-V1-PE-HMB-001 (design case unless marked), mechanical data from the equipment list CHX-V1-ME-EQL-001, motor ratings from CHX-V1-EE-LL-001.
• Nozzle schedules: size from the line list CHX-V1-PE-LL-001, flange class from ASME B16.5 group 1.1 at the design temperature; the EPC P&ID line (CHX-V1-PID-001) shown beside it, ≠ where its size differs.
• R-101 carries its GA sketch. Every number from the model or the other documents, or labelled assumed / typical practice.

Revisions

Rev	Date	Description	Originator	Checker	Approver
0	2026-10-26	Issued for teaching	Apinan Soottitantawat		

This document is generated for teaching from the simulator model of the unit. It is not issued for construction; every number comes from the model or is labelled assumed / typical practice. No client, contractor or vendor is implied.

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CHX-V1-ME-DS-R-101 · Main hydrogenation reactor with agitator M-101

Item	Value
Tag / quantity	R-101 (1)
Service	Benzene + H_2 → cyclohexane on Raney-Ni slurry; pump-around cooling through E-101, boil-up to E-103
Type	Vertical agitated pressure vessel, 2:1 ellipsoidal heads, skirt
Size / capacity	ID 3,800 × T/T 4,500 mm · 65.4 m^3 total; 29.0 m^3 liquid at NLL (LI-101 65 %)
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-01 (CHX-V1-PC-PKG-001)
Driver	M-101: 250 kW, 6.6 kV

Process data

Quantity	Value	Source
Liquid (slurry) at NLL	29.0 m^3 clear, 32.5 m^3 aerated (gas hold-up 12 %)	model n_1/ρ, LI-101
Operating	30.00 bar g · 200.0 °C (TIC-101)	model
Benzene feed (N1)	34.80 t/h · 40.0 °C · 30.00 bar g	HMB D
Hydrogen to the sparger (N2)	6.42 t/h · 60.0 °C · 32.00 bar g · 46,000 Nm^3/h	HMB D
Overhead to E-103 (N3)	51.19 t/h · 200.0 °C · 30.00 bar g	HMB D
Reflux (N4)	46.46 t/h · 55.7 °C · 30.00 bar g	HMB D
Pump-around out / return (N5/N6)	606 t/h out; return 357 t/h at 159.3 °C + 248 t/h by-pass	HMB D
Slurry to R-102 (N7)	36.49 t/h · 200.0 °C · 30.00 bar g	HMB D
Heat of reaction released	24.98 MW	HMB D
Boiling ceiling TbC(PI-101)	258.3 °C; 272.4 °C at the PSV set	model TbC
Hydrogen partial pressure	13.76 bar	Design tab

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	SL_ASSUMED.Pdes = PSV-101 set
Design temperature	250 °C	SL_ASSUMED.Tdes1 (critical limit of the R-101 ladder)
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	53.4 / 55 mm	$t = PR/(SE - 0.6P) + CA$
Heads 2:1: required / plate	52.8 / 60 mm	$t = PD/(2SE - 0.2P) + CA$, +10 % forming
MAWP (corroded, hot)	36.9 bar g	from the plates chosen
Hydrotest (UG-99(b))	46.8 bar g at the top	1.3 × design × S(test)/S(design)
Levels (above the vessel bottom)	LLL 2.30 · NLL 3.18 · HLL 3.93 · LI-101 100 % 4.73 m; TTL 5.45 m	LI-101 linearised to volume (strapping table): 0–100 % = 0–50 m^3 as in the model
Internals	4 baffles (T/12), H_2 sparger ring below the bottom impeller, steady bearing (none: cantilever shaft, assumed)	typical
Material	SA-516 Gr 70 N + PWHT (caustic residue of Raney-Ni, hot H_2); internals 316L	typical
Weight (estimate)	empty 49.0 t; operating 66.6 t	steel × 1.25 (nozzles, internals, skirt); drive under M-101 (assumed)

Agitator M-101

Quantity	Value	Basis
Impellers	bottom: Concave-blade gas-dispersing disc turbine Ø1.52 m (Np 3.2); upper: Hydrofoil (axial) Ø1.71 m (Np 0.3)	typical choice for gas dispersion + top-to-bottom circulation
Speed	128 rpm (VSD); tip speed 10.2 m/s	from $P = Np\ \rho\ N^3\ D^5$ at the model ungassed power
Power	ungassed 180 kW (model K.agitKW); gassed 153 kW (JI-101), ratio 0.85	model
Gas flow number / flooding limit	Fl 0.096 vs 2.15 → not flooded	typical correlation
Motor	250 kW, 6.6 kV, VSD (model maximum 204.3 kW)	CHX-V1-EE-LL-001
Seal	double mechanical seal, pressurised barrier fluid (M-101-AUX)	typical for H_2

Nozzle schedule

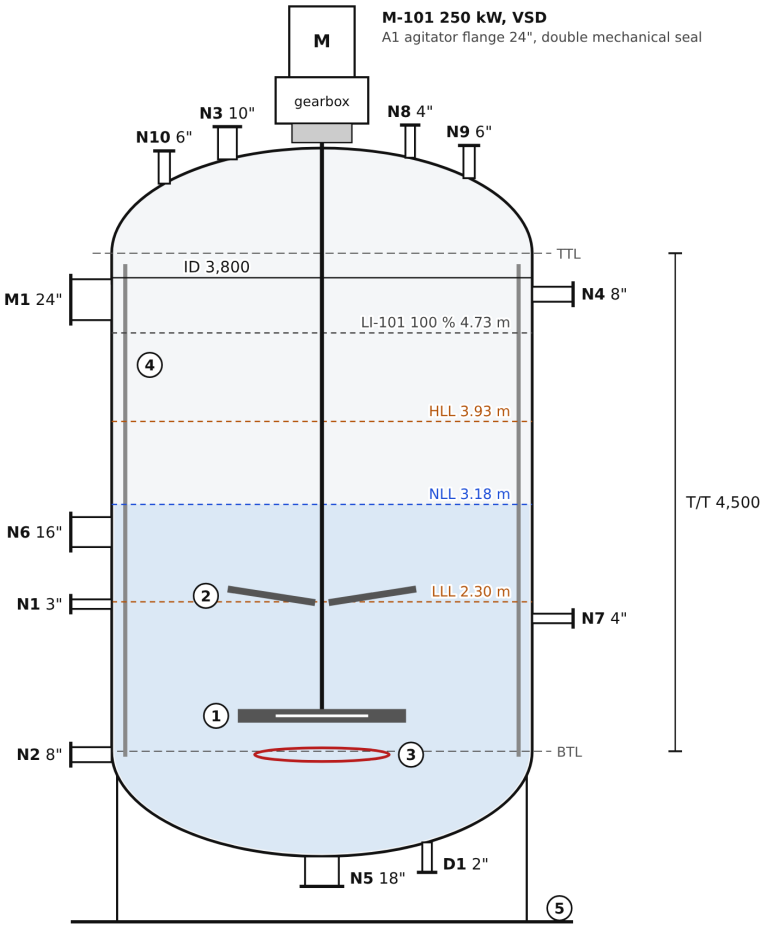
Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Benzene feed (FV-101, XV-101)	3"	300 RF	3"-P-1002-B1	4"-P-1002 ≠	36 bar g / 250 °C
N2	Hydrogen to the sparger (FV-103)	8"	300 RF	8"-H-2009-B2	8"-G-1009	36 bar g / 250 °C
N3	Overhead gas + vapour to E-103	10"	300 RF	10"-PG-2001-B2	10"-P-1005	36 bar g / 250 °C
N4	Reflux from V-103	8"	300 RF	8"-P-1009-B1	6"-P-1007 ≠	36 bar g / 250 °C
N5	Pump-around outlet to P-101A/B	18"	300 RF	18"-P-1003-B3	18"-SL-1010	36 bar g / 250 °C
N6	Pump-around return (E-101 + FV-104 by-pass)	16"	300 RF	16"-P-1006-B3	16"-SL-1016	36 bar g / 250 °C
N7	Slurry to R-102 (LV-101)	4"	300 RF	4"-P-1008-B3	6"-SL-1014 ≠	36 bar g / 250 °C
N8	Catalyst charge from D-102 (HV-117)	4"	300 RF	—	4"-SL-1004	36 bar g / 250 °C
N9	Start-up cyclohexane (HV-116)	6"	300 RF	—	6"-P-1003	36 bar g / 250 °C
N10	PSV-101 inlet	6"	300 RF	—	6"-FL-1040	36 bar g / 250 °C
A1	Agitator M-101 mounting flange	24"	300 RF	—	—	36 bar g / 250 °C
M1	Manway	24"	300 RF	—	—	36 bar g / 250 °C
K1	LT-101 lower tap / LSLL-101	2"	300 RF	—	—	36 bar g / 250 °C
K2	LT-101 upper tap	2"	300 RF	—	—	36 bar g / 250 °C
K3	TE-101 / TE-100 thermowells (separate BPCS and SIS)	1-1/2"	300 RF	—	—	36 bar g / 250 °C
K4	PT-101 and PSHH-101 (separate taps)	2"	300 RF	—	—	36 bar g / 250 °C
D1	Drain (to closed drain, flushing)	2"	300 RF	—	—	36 bar g / 250 °C

Note: Nozzle sizes follow the line list CHX-V1-PE-LL-001; where the EPC P&ID draws another size the P&ID is to be revised (EQ-em-4).
Note: Design temperature 250 °C is below the boiling ceiling (EQ-em-1).

General arrangement (sketch)

R-101 · general arrangement (elevation, sketch)

ID 3,800 × T/T 4,500 mm · 2:1 ellipsoidal heads · design 36 bar g / 250 °C · SA-516 Gr 70 N + PWHT, shell 55 mm, heads 60 mm



Internals

1. gas-dispersing turbine Ø1.52 m
2. upper axial (hydrofoil) impeller
3. H_2 sparger ring (N2)
4. four baffles, T/12 wide
5. skirt on a concrete plinth (assumed)

Dimensions in mm; levels in m above the bottom of the lower head.
Nozzle orientation is schematic (all nozzles shown in one plane); sizes and services in the nozzle schedule.
LI-101 is linearised to volume (0–100 % = 0–50 m³). Aerated liquid 32.5 m³ at NLL = 29.0 m³ clear (model).

CHX-V1-ME-DS-R-101 · GA
Cyclohexane Reactor Ver 1 · Rev 0
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CHX-V1-ME-DS-R-102 · Finishing slurry reactor (jacketed) with agitator M-102 (HOLD)

Item	Value
Tag / quantity	R-102 (1)
Service	Converts the remaining benzene (≈2 % → ≈100 ppm); jacket on MP steam (HEAT) or CW (COOL) by HS-102
Type	Vertical agitated pressure vessel with jacket, 2:1 heads, skirt
Size / capacity	ID 3,200 × T/T 3,800 mm · 39.1 m³ total; 15.0 m³ liquid at NLL (LI-102 55 %)
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-01 (CHX-V1-PC-PKG-001)
Driver	M-102: HOLD

Process data

Quantity	Value	Source
Liquid at NLL	15.0 m³ clear, 16.5 m³ aerated	model
Operating	29.60 bar g · 160.0 °C (TIC-102)	model
Slurry from R-101 (N1)	36.49 t/h · 200.0 °C · 30.00 bar g	HMB D
Gas from V-103 (N2)	4.73 t/h · 55.7 °C · 29.60 bar g	HMB D
Gas + vapour to R-103 (N3)	20.28 t/h · 160.0 °C · 29.60 bar g	HMB D
Bottom to P-102 (N4)	20.94 t/h · 160.0 °C · 29.60 bar g	HMB D
Jacket duty	+1.29 MW heating on MP steam (2.33 t/h); capacity 2.16 MW	model
Benzene in → out	2.00 % → 102 ppm	model

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	SL_ASSUMED.Pdes = PSV-102 set
Design temperature	250 °C	as R-101 (SL_ASSUMED.Tdes1); covers MP steam 184 °C
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	45.5 / 50 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	44.9 / 50 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	36.2 bar g	from the plates chosen
Hydrotest (UG-99(b))	46.8 bar g at the top	1.3 × design × S(test)/S(design)
Jacket	conventional jacket on shell and bottom head; design 12 bar g / 200 °C (EQ-ep-4); TSV-102J	EQ-ep-4 resolution
Jacket wetted area / U needed	26.4 m² → U = UAj/A = 3.41 kW/m²K for the model UAj 90 kW/K	typical jacket U 0.3–0.6 (EQ-em-9)
Levels	NLL 2.32 m; LI-102 100 % 4.00 m above the bottom	linearised as R-101
Agitator M-102	HOLD — ≈ 23 kW (1.5 kW/m³ typical, assumed) → 30 kW motor	EQ-em-7
Weight (estimate)	empty 31.8 t; operating 41.4 t	steel × 1.30 (jacket, nozzles, skirt); drive under M-102 (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Slurry from R-101 (LV-101)	4"	300 RF	4"-P-1008-B3	6"-SL-1014 ≠	36 bar g / 250 °C
N2	Gas from V-103 to the sparger	6"	300 RF	6"-PG-2003-B2	6"-G-1008	36 bar g / 250 °C
N3	Gas + vapour to R-103	8"	300 RF	8"-PG-2004-B2	8"-P-2004	36 bar g / 250 °C
N4	Bottom outlet to P-102A/B	4"	300 RF	4"-P-1010-B3	6"-SL-2001 ≠	36 bar g / 250 °C
N5	Off-gas to EDV-101 / HV-121 (upstream of the valves)	4"	300 RF	—	4"-FL-2015	36 bar g / 250 °C
N6	PSV-102 inlet	8"	300 RF	—	8"-FL-2016	36 bar g / 250 °C
J1	Jacket MP steam in (HS-102 HEAT, TV-102)	6"	150 RF	6"-S-3003-A1	4"-MS-2020 ≠	12 bar g / 200 °C
J2	Jacket CW in (HS-102 COOL)	6"	150 RF	6"-CW-3011-A1	6"-CW-2021	12 bar g / 200 °C
J3	Jacket CW out	6"	150 RF	—	6"-CW-2022	12 bar g / 200 °C
J4	Jacket condensate (ST-102)	2"	150 RF	—	2"-CD-2023	12 bar g / 200 °C
A1	Agitator M-102 mounting flange (HOLD)	16"	300 RF	—	—	36 bar g / 250 °C
M1	Manway	24"	300 RF	—	—	36 bar g / 250 °C

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
K1	LT-102 taps (2)	2"	300 RF	—	—	36 bar g / 250 °C
K2	TE-102 thermowell	1-1/2"	300 RF	—	—	36 bar g / 250 °C
K3	PT-102	2"	300 RF	—	—	36 bar g / 250 °C
D1	Drain	2"	300 RF	—	—	36 bar g / 250 °C

Note: The jacket and the CW branch up to the HS-102 change-over valves are 12 bar g / 200 °C (EQ-ep-4).

CHX-V1-ME-DS-R-103 · Trickle-bed finishing reactor

Item	Value
Tag / quantity	R-103 (1)
Service	Adiabatic polishing of benzene to ≈2 ppm on supported Ni; liquid from F-102/LV-102 and gas from R-102 in co-current down-flow
Type	Vertical fixed-bed vessel, 2:1 heads, skirt; distributor tray, support grid, inert balls
Size / capacity	ID 1,950 × T/T 7,400 mm; bed 5.71 m · 16.2 m³ catalyst (LHSV 3 h^{-1}, Design tab)
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-02 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Gas in (N1)	20.28 t/h · 160.0 °C · 29.60 bar g	HMB D
Liquid in (N2)	20.94 t/h · 160.0 °C · 29.25 bar g	HMB D
Effluent (N3)	41.22 t/h · 158.6 °C · 29.25 bar g	HMB D
Liquid to the bed	48.7 m³/h at 15 °C; G_L 3.7 kg/m²·s	Design tab
Bed temperature rise	0.08 K at design; 8.3 K per 1 % benzene slip	model / Design tab
Benzene in → out	102 → 2.0 ppm	model

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	SL_ASSUMED.Pdes = PSV-102 set (no isolation from R-102)
Design temperature	260 °C	SL_ASSUMED.Tdes3 (critical limit of the bed ladder)
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	28.9 / 30 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	28.6 / 32 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	36.3 bar g	from the plates chosen
Hydrotest (UG-99(b))	46.8 bar g at the top	1.3 × design × S(test)/S(design)
Catalyst	16.2 m³ supported Ni, bed 5.71 m; ≈ 14.6 t at 900 kg/m³ (assumed)	Design tab LHSV 3 h^{-1}
Internals	inlet diffuser, distributor tray (≥ 100 drip points/m², typical), 0.3 m inert balls top and bottom, support grid, outlet collector	typical
Thermocouples	TE-107A…E multipoint at five bed elevations (TSHH-107 on its own element, SL)	P&ID
Weight (estimate)	empty 17.0 t; operating 34.6 t	steel × 1.30 (internals); + 14.6 t catalyst at 900 kg/m³ and 3 t liquid hold-up (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Gas inlet from R-102 (top distributor)	8"	300 RF	8"-PG-2004-B2	8"-P-2004	36 bar g / 260 °C
N2	Liquid inlet from F-102 / LV-102 (top distributor)	3"	300 RF	3"-P-1012-B1	3"-P-2003	36 bar g / 260 °C
N3	Effluent to E-106	8"	300 RF	8"-PG-2005-B2	10"-P-2005 ≠	36 bar g / 260 °C
M1	Manway / catalyst loading (top)	24"	300 RF	—	—	36 bar g / 260 °C
N4	Catalyst unloading (bottom, 45°)	6"	300 RF	—	—	36 bar g / 260 °C
K1	TE-107A…E multipoint thermowell (5 cells)	3"	300 RF	—	—	36 bar g / 260 °C
K2	PDT across the bed (assumed)	1"	300 RF	—	—	36 bar g / 260 °C
D1	Drain	2"	300 RF	—	—	36 bar g / 260 °C

CHX-V1-ME-DS-E-101 · Pump-around cooler / LP steam generator

Item	Value
Tag / quantity	E-101 (1)
Service	Tubes: R-101 slurry 200 → 159.3 °C; shell: boiling BFW at the D-101 pressure (COOL) or MP steam (HEAT, start-up)
Type	Shell and tube, TEMA AEL, horizontal; 4 tube passes; natural circulation (thermosyphon) to D-101
Size / capacity	Shell ID 1,250 mm × 1210 tubes Ø25.4 × 7 320 mm · 9.60 MW (model); A 707 m²
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-03 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Duty	9,598 kW (120 %: 11,691; 70 %: 6,349)	model P.Qe1 / HMB
Tube side: slurry	357 t/h (FIC-104) · 200.0 → 159.3 °C · 32.00 bar g	HMB 10/11
Shell side: boiling water	155.8 °C at 4.50 bar g (PIC-110); steam 14.72 t/h; circulation ratio 8 (assumed)	model
UA / NTU / ε (model relation)	600 kW/K / 2.54 / 0.921	UA = $UA_1(F/1.5)^{0.3}$
LMTD	16.0 K	(T_1−Tsat) and (T_out−Tsat)
Maximum duty at full pump flow	14,619 kW (1.52 × design)	clean
Fouling margin	UA may fall to 45 % of clean (R_f 1.25 m²K/kW) before the design duty needs the full P-101 flow; 62 % at 120 %	design margin
HEAT mode	MP steam 184 °C on the shell (HV-110), ≤ 15.1 t/h	model

Mechanical data

Quantity	Value	Basis
Design pressure	40.0 bar g (other side 12.0 bar g)	tubes: R-101 PSV 36 + P-101 shut-off rise 4 bar (as line 16"-P-1005-B3); shell: MP steam line 12 bar g (8"-S-3002-A1)
Design temperature	250 °C (other side 200 °C)	tubes SL_ASSUMED.Tdes1; shell = MP steam line design 200 °C
TEMA type / passes	AEL, 4 tube passes, horizontal	typical
Tubes	1210 × Ø25.4 × 2.11 mm × 7 320 mm, 316L, triangular pitch 31.75 mm; area 707 m² (Design tab 706 m² at U 0.85)	Design tab area
Shell ID / bundle	1,250 / 1,217 mm	bundle correlation, typical
Tube velocity	1.49 m/s design; 2.53 at full pump flow; 0.94 at 70 %	slurry 1–3 m/s
Tube-side ΔP	0.33 bar at design (allowable 0.64 bar, Process engineering); 0.85 bar at full flow	Blasius + 4 velocity heads per pass (typical)
Shell / channel thickness	shell 10 mm (12 bar g, E 0.85); channel 22 mm (40 bar g)	UG-27
Fixed tubesheets	mean metal ΔT shell–tubes ≈ 24 K ≤ 50 K → no expansion joint (typical)	check
Weight (estimate)	empty 19.3 t; operating 26.1 t	tubes + shell + tubesheets + channels × 1.15; operating +35 % for the water and slurry (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Tube inlet: slurry from FV-104	16"	300 RF	16"-P-1005-B3	12"-SL-1015 ≠	40 bar g / 250 °C
N2	Tube outlet: slurry to R-101	16"	300 RF	16"-P-1006-B3	12"-SL-1012 ≠	40 bar g / 250 °C
N3	Shell inlet: downcomer from D-101 (HY-101A)	8"	150 RF	8"-BFW-3005-A1	12"-BW-1022 ≠	12 bar g / 200 °C
N4	Shell outlet: riser to D-101 (HY-101B)	16"	150 RF	16"-S-3006-A1	14"-LS-1023 ≠	12 bar g / 200 °C
N5	Shell: MP steam in (HS-101 HEAT)	8"	150 RF	8"-S-3002-A1	8"-MS-1024	12 bar g / 200 °C
N6	Shell: condensate (HEAT, ST-101)	3"	150 RF	—	3"-CD-1025	12 bar g / 200 °C
K1	TE-104 (tube outlet, on the line)	1-1/2"	300 RF	—	—	40 bar g / 250 °C
V1	Shell vent	1"	150 RF	—	—	12 bar g / 200 °C
D1	Shell drain	2"	150 RF	—	—	12 bar g / 200 °C
D2	Channel drains / flushing (2)	2"	300 RF	—	—	40 bar g / 250 °C

Note: Shell side is designed for MP steam (12 bar g) because HY-101A/B isolate it from D-101 in HEAT; PSV-109 protects it only in COOL (EQ-em-2).
Note: Tube rupture: the shell (12 bar g) is below 10/13 of the tube side (40 bar g) — a relief case for D-101 (EQ-em-3).

CHX-V1-ME-DS-E-103 · Reflux condenser (cooling water)

Item	Value
Tag / quantity	E-103 (1)
Service	Condenses the cyclohexane boiled up in R-101 out of the hydrogen; condensate refluxes via V-103
Type	Shell and tube, TEMA AES, horizontal; CW in 2-pass tubes, process vapour on the shell
Size / capacity	Shell ID 950 mm × 991 tubes Ø19.05 × 6 100 mm · 7.87 MW (HMB); A 362 m²
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-03 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Duty	7,872 kW (120 %: 9,358)	HMB
Shell side: process	51.19 t/h · 200.0 °C · 30.00 bar g → 55.7 °C, vapour fraction 0.58 out	HMB 4/5
Tube side: cooling water	432 t/h · 32 → 47.7 °C (120 %: 50.7 °C)	model K.Fcw3
LMTD × F	69.1 K × 0.896 (1 shell, 2 tube passes)	HMB D
U (assumed) / UA required	0.45 kW/m²K / 127 kW/K	typical for H_2-rich partial condensation / cooling
Area required / installed	282 / 362 m²	larger of 1.2 × design and 1.1 × the 120 % case
Rated capacity (U × A × F × LMTD)	10,086 kW design; 10,296 kW at 120 % conditions	HMB D
Model condenser	UA_3 500 kW/K (model) vs 163 kW/K here; model maximum 47.9 MW	EQ-em-9

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g (other side 10.0 bar g)	shell SL_ASSUMED.Pdes (PSV-101, no isolation); tubes CW 10 bar g (TSV set)
Design temperature	250 °C (other side 65 °C)	shell SL_ASSUMED.Tdes1; tubes CW supply line 65 °C (LL)
TEMA type / passes	AES, 2 tube passes, horizontal	floating head: shell–tube ΔT large
Tubes	991 × Ø19.05 × 1.65 mm × 6 100 mm, CS; area 362 m²	
Shell ID	950 mm	
CW velocity in tubes	1.25 m/s	1–2.5 m/s typical
Shell thickness	16 mm	UG-27
Weight (estimate)	empty 8.9 t; operating 11.6 t	tubes + shell × 1.3 (heads, floating head) × 1.15; operating +30 % (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Shell inlet: R-101 overhead	10"	300 RF	10"-PG-2001-B2	10"-P-1005	36 bar g / 250 °C
N2	Shell outlet: to V-103	8"	300 RF	8"-PG-2002-B2	6"-P-1006 ≠	36 bar g / 250 °C
N3	Tube inlet: CW supply (HV-113)	10"	150 RF	10"-CW-3007-A1	10"-CW-1030	10 bar g / 65 °C
N4	Tube outlet: CW return	10"	150 RF	10"-CW-3008-A1	10"-CW-1031	10 bar g / 65 °C
K1	TSV-113 (CW side)	1"	150 RF	—	—	10 bar g / 65 °C
V1	Vents (2)	1"	300 RF	—	—	36 bar g / 250 °C
D1	Drains (2)	2"	300 RF	—	—	36 bar g / 250 °C

Note: Tube rupture: the CW side (10 bar g) is below 10/13 of the shell (36 bar g) — hydrogen into the CW system is a relief case (EQ-em-3).

CHX-V1-ME-DS-E-106 · Product cooler (cooling water)

Item	Value
Tag / quantity	E-106 (1)
Service	Cools the R-103 effluent (gas + liquid) to the V-104 temperature
Type	Shell and tube, TEMA AES, horizontal; CW in 6-pass tubes
Size / capacity	Shell ID 950 mm × 936 tubes Ø19.05 × 6 100 mm · 3.49 MW (HMB, sensible estimate); A 342 m²
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-03 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Duty	3,492 kW (120 %: 4,192)	HMB
Shell side: process	41.22 t/h · 158.6 °C · 29.25 bar g → 44.9 °C, vapour fraction 0.62 out	HMB 16/17
Tube side: cooling water	216 t/h · 32 → 45.9 °C (120 %: 48.7 °C)	econ() 60 kg/s (assumed)
LMTD × F	46.0 K × 0.808 (1 shell, 6 tube passes)	HMB D
U (assumed) / UA required	0.4 kW/m²K / 94 kW/K	typical for H_2-rich partial condensation / cooling
Area required / installed	235 / 342 m²	larger of 1.2 × design and 1.1 × the 120 % case
Rated capacity (U × A × F × LMTD)	5,085 kW design; 4,615 kW at 120 % conditions	HMB D
Model cooler	fixed ε 0.898 on (T_in − T_cw) (model)	model

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g (other side 10.0 bar g)	shell SL_ASSUMED.Pdes (PSV-104); tubes CW 10 bar g
Design temperature	260 °C (other side 65 °C)	shell = R-103 outlet line (SL_ASSUMED.Tdes3); tubes 65 °C
TEMA type / passes	AES, 6 tube passes, horizontal	floating head: shell–tube ΔT large
Tubes	936 × Ø19.05 × 1.65 mm × 6 100 mm, CS; area 342 m²	
Shell ID	950 mm	
CW velocity in tubes	1.98 m/s	1–2.5 m/s typical
Shell thickness	16 mm	UG-27
Weight (estimate)	empty 8.6 t; operating 11.2 t	as E-103 (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Shell inlet: R-103 effluent	8"	300 RF	8"-PG-2005-B2	10"-P-2005 ≠	36 bar g / 260 °C
N2	Shell outlet: to V-104	8"	300 RF	8"-PG-2006-B2	8"-P-2006	36 bar g / 260 °C
N3	Tube inlet: CW supply	8"	150 RF	8"-CW-3009-A1	10"-CW-2030 ≠	10 bar g / 65 °C
N4	Tube outlet: CW return	8"	150 RF	8"-CW-3010-A1	10"-CW-2030 ≠	10 bar g / 65 °C
K1	TSV-106 (CW side)	1"	150 RF	—	—	10 bar g / 65 °C
V1	Vents (2)	1"	300 RF	—	—	36 bar g / 260 °C
D1	Drains (2)	2"	300 RF	—	—	36 bar g / 260 °C

Note: Tube rupture: the CW side (10 bar g) is below 10/13 of the shell (36 bar g) — hydrogen into the CW system is a relief case (EQ-em-3).

CHX-V1-ME-DS-D-101 · Steam drum

Item	Value
Tag / quantity	D-101 (1)
Service	Raises LP steam from E-101 by natural circulation; PIC-110 / PV-110 to the LP header
Type	Horizontal drum, 2:1 heads, saddles; steam separators and demister
Size / capacity	ID 2,400 × T/T 8,400 mm · water span 24 t (model K.Mspan) = 24.9 m³ between LT-109 0 and 100 %
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-04 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Operating	4.50 bar g · 155.8 °C (PIC-110); level 50 % (LIC-109)	model
LP steam out (N2)	14.72 t/h · 155.8 °C · 4.50 bar g	HMB D
BFW in (N1)	14.87 t/h · 105.0 °C · 9.99 bar g	HMB D
Blowdown (N5)	0.15 t/h · 155.8 °C · 4.50 bar g	HMB D
Water span	24 t (K.Mspan) = 24.9 m³ at 964 kg/m³	model

Mechanical data

Quantity	Value	Basis
Design pressure	8.0 bar g	SL_ASSUMED.Psv109 = PSV-109 set
Design temperature	180 °C	saturation at the set + margin (as 10"-S-3001-A1)
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	0.85 / 1.5 mm	assumed
Shell: required / plate	9.7 / 12 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	9.7 / 12 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	8.9 bar g	from the plates chosen
Hydrotest (UG-99(b))	10.4 bar g at the top	1.3 × design × S(test)/S(design)
Level geometry	LT-109 0 % at 20 % of D, 100 % at 70 % of D: span 25.5 m³	assumed taps
Internals	cyclone/chevron steam separators, demister, BFW distributor, blowdown collector	typical
Weight (estimate)	empty 8.6 t; operating 26.1 t	steel × 1.20 (internals, saddles) + water at NWL (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	BFW inlet (LV-109) to the internal distributor	3"	150 RF	3"-BFW-3004-A1	3"-BW-1020	8 bar g / 180 °C
N2	LP steam outlet (PV-110)	10"	150 RF	10"-S-3001-A1	10"-LS-1021	8 bar g / 180 °C
N3	Downcomer to E-101	8"	150 RF	8"-BFW-3005-A1	12"-BW-1022 ≠	8 bar g / 180 °C
N4	Riser return from E-101	16"	150 RF	16"-S-3006-A1	14"-LS-1023 ≠	8 bar g / 180 °C
N5	Continuous blowdown	1"	150 RF	—	1"-BD-1026	8 bar g / 180 °C
N6	PSV-109 inlet	4"	150 RF	—	—	8 bar g / 180 °C
K1	LT-109 bridle (2)	2"	150 RF	—	—	8 bar g / 180 °C
K2	LSLL-109 (own taps)	2"	150 RF	—	—	8 bar g / 180 °C
K3	PT-110	1"	150 RF	—	—	8 bar g / 180 °C
K4	Level gauge (2)	1"	150 RF	—	—	8 bar g / 180 °C
M1	Manway	20"	150 RF	—	—	8 bar g / 180 °C
D1	Intermittent blowdown / drain	2"	150 RF	—	—	8 bar g / 180 °C

Note: D-101 stays 8.0 bar g with PSV-109 (SL_ASSUMED); MP steam reaches it only if HY-101A/B fail to close in HEAT — relief case for the Relief & vent tab (EQ-em-2).

CHX-V1-ME-DS-D-102 · Catalyst charge pot

Item	Value
Tag / quantity	D-102 (1)
Service	Holds a Raney-Ni charge under liquid and N_2; charged to R-101 through HV-117
Type	Vertical pot, 2:1 heads, conical-bottom option, legs (assumed size)
Size / capacity	ID 1,000 × T/T 1,500 mm (assumed) · 1.44 m^3 (one make-up batch, assumed)
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-01 (CHX-V1-PC-PKG-001)
HOLD	Charging method and pressure source to confirm (EQ-em-8).

Process data

Quantity	Value	Source
Content	Raney-Ni slurry in cyclohexane under N_2 (pyrophoric when dry)	P&ID
Batch	≈ 0.9 m^3 slurry (60 % full, assumed); catalyst inventory not in the model	assumed
Operating	ambient, N_2 blanket; transfers to R-101 through HV-117	P&ID

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	as R-101 (connected through HV-117)
Design temperature	250 °C	as R-101
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-240 316L (stainless plate) · 103 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 0 mm	assumed
Shell: required / plate	17.9 / 20 mm	$t = PR/(SE - 0.6P) + CA$
Heads 2:1: required / plate	17.5 / 20 mm	$t = PD/(2SE - 0.2P) + CA$, +10 % forming
MAWP (corroded, hot)	36.8 bar g	from the plates chosen
Hydrotest (UG-99(b))	52.3 bar g at the top	1.3 × design × S(test)/S(design)
Weight (estimate)	empty 1.5 t; operating 3.8 t	steel × 1.3 + slurry charge (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Catalyst slurry outlet to R-101 (HV-117)	4"	300 RF	—	4"-SL-1004	36 bar g / 250 °C
N2	N_2 pressurising / blanket	1"	300 RF	—	—	36 bar g / 250 °C
N3	Vent to flare	2"	300 RF	—	—	36 bar g / 250 °C
N4	Flushing cyclohexane	2"	300 RF	—	—	36 bar g / 250 °C
M1	Charging hatch (drums)	12"	300 RF	—	—	36 bar g / 250 °C
K1	PG / PT	1"	300 RF	—	—	36 bar g / 250 °C
K2	LG	1"	300 RF	—	—	36 bar g / 250 °C

Note: The N_2 header (8 bar g) cannot push a charge into R-101 at 30 bar g: define the charging method (EQ-em-8).

CHX-V1-ME-DS-V-103 · Reflux drum

Item	Value
Tag / quantity	V-103 (1)
Service	Separates the E-103 condensate (reflux to R-101 by gravity) from the hydrogen to R-102
Type	Vertical drum, 2:1 heads, mesh demister, skirt
Size / capacity	ID 1,200 × T/T 3,000 mm (PFD size) · 3.8 m³; gas 0.20 m/s vs 0.68 allowed (120 %)
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-05 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Inlet (N1)	51.19 t/h · 55.7 °C · 30.00 bar g, vapour fraction 0.58	HMB D
Gas to R-102 (N2)	4.73 t/h · 55.7 °C · 29.60 bar g	HMB D
Reflux to R-101 (N3)	46.46 t/h · 55.7 °C · 30.00 bar g	HMB D
Gas velocity / allowed (120 %)	0.20 / 0.68 m/s (K = 0.07 m/s with mesh, typical)	Souders–Brown
Liquid residence half full	1.5 min at the 120 % reflux	gravity reflux, no level control in the model

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	SL_ASSUMED.Pdes (PSV-101, no isolation)
Design temperature	250 °C	as the E-103 outlet line (loss of cooling)
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	19.0 / 20 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	18.8 / 22 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	37.6 bar g	from the plates chosen
Hydrotest (UG-99(b))	46.8 bar g at the top	1.3 × design × S(test)/S(design)
Weight (estimate)	empty 2.9 t; operating 4.3 t	steel × 1.2 + liquid half full (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Inlet from E-103	8"	300 RF	8"-PG-2002-B2	6"-P-1006 ≠	36 bar g / 250 °C
N2	Gas outlet to R-102	6"	300 RF	6"-PG-2003-B2	6"-G-1008	36 bar g / 250 °C
N3	Reflux outlet to R-101	8"	300 RF	8"-P-1009-B1	6"-P-1007 ≠	36 bar g / 250 °C
K1	LG-103 (2)	2"	300 RF	—	—	36 bar g / 250 °C
K2	TE-103 (vent gas)	1-1/2"	300 RF	—	—	36 bar g / 250 °C
M1	Manway	18"	300 RF	—	—	36 bar g / 250 °C
D1	Drain	2"	300 RF	—	—	36 bar g / 250 °C

CHX-V1-ME-DS-V-104 · HP separator

Item	Value
Tag / quantity	V-104 (1)
Service	Separates the product liquid from the recycle hydrogen at 29 bar g; make-up H_2 enters here (PFD)
Type	Horizontal drum, 2:1 heads, saddles; inlet device, mesh demister, vortex breaker
Size / capacity	ID 2,300 × T/T 9,100 mm · level span 31.0 m^3 (model K.V4span 30 m^3)
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-05 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Inlet (N1)	41.22 t/h · 44.9 °C · 29.00 bar g, vapour fraction 0.62	HMB D
Gas to K-101 (N2)	6.42 t/h · 44.9 °C · 29.00 bar g	HMB D
Product (N3)	37.50 t/h · 44.9 °C · 29.00 bar g	HMB D
Purge (N4)	0.28 t/h · 44.9 °C · 29.00 bar g	HMB D
Make-up H_2 (N5)	2.98 t/h · 40.0 °C · 34.99 bar g	HMB D
Operating	29.00 bar g (PIC-105) · 44.9 °C; level 50 % (LIC-104)	model
Gas velocity above LSHH / allowed	0.537 / 1.68 m/s (horizontal, K 0.12 m/s typical)	Souders–Brown, 120 % gas
Liquid residence (span / product)	36 min	model span

Mechanical data

Quantity	Value	Basis
Design pressure	36.0 bar g	SL_ASSUMED.Pdes = PSV-104 set
Design temperature	190 °C	R-103 outlet + 30 °C (loss of E-106 cooling, typical rule)
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	33.6 / 36 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	33.2 / 40 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	38.5 bar g	from the plates chosen
Hydrotest (UG-99(b))	46.8 bar g at the top	1.3 × design × S(test)/S(design)
Level geometry	LT-104 0 % at 10 % of D, 100 % at 75 % of D: span 31.0 m^3; LSHH-104 90 % at 1.58 m	assumed taps
Weight (estimate)	empty 27.3 t; operating 39.6 t	steel × 1.2 + liquid at NLL (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Inlet from E-106	8"	300 RF	8"-PG-2006-B2	8"-P-2006	36 bar g / 190 °C
N2	Gas outlet to K-101	8"	300 RF	8"-H-2007-B2	10"-G-2007 ≠	36 bar g / 190 °C
N3	Product outlet (LV-104)	4"	300 RF	4"-P-1013-B1	6"-P-2011 ≠	36 bar g / 190 °C
N4	Purge to fuel gas (HIC-107)	1-1/2"	300 RF	1-1/2"-H-2011-B2	2"-G-2010 ≠	36 bar g / 190 °C
N5	Make-up H_2 (PFD/LL; P&ID ties in at the K-101 suction, EQ-px-3)	6"	300 RF	6"-H-2010-B1	8"-G-2009 ≠	36 bar g / 190 °C
N6	PSV-104 inlet	8"	300 RF	—	8"-FL-2016	36 bar g / 190 °C
K1	LT-104 / LT-104S taps (LIC-104, LSHH-104, LSLL-104)	2"	300 RF	—	—	36 bar g / 190 °C
K2	TE-109	1-1/2"	300 RF	—	—	36 bar g / 190 °C
K3	PT-105 (PIC-105)	2"	300 RF	—	—	36 bar g / 190 °C
M1	Manway	24"	300 RF	—	—	36 bar g / 190 °C
D1	Drain / water boot	2"	300 RF	—	—	36 bar g / 190 °C

Note: Design temperature covers loss of E-106 cooling (EQ-ep-12); the outlet lines in the line list are 80 °C (EQ-em-10).
Note: PFD draws a vertical 2.0 × 5.0 m vessel; the model level span needs the horizontal drum given here (EQ-em-6).

CHX-V1-ME-DS-F-102 · Catalyst candle filters F-102A/B

Item	Value
Tag / quantity	F-102A/B (2 × 100 %)
Service	Hold back the Raney-Ni from the R-102 liquid before R-103; PDI-102
Type	Vertical filter housing, bolted top cover, sintered-metal candles, back-flush (assumed)
Size / capacity	ID 800 × T/T 2,400 mm; 48 candles Ø60 × 1500 mm (assumed) · 13.6 m²; 2.41 m³/m²·h at design
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-08 (CHX-V1-PC-PKG-001)

Process data

Quantity	Value	Source
Inlet (N1)	20.94 t/h · 160.0 °C · 32.60 bar g	HMB D
Filtrate (N2)	20.94 t/h · 160.0 °C · 29.25 bar g	HMB D
Flow	32.6 m³/h (120 %: 39.5)	HMB
Clean ΔP / alarm	0.26 bar (PDI-102) / 0.8 bar	model / ALM
Flux	2.41 m³/m²·h (typical 1–3)	assumed area

Mechanical data

Quantity	Value	Basis
Design pressure	40.0 bar g	P-102 discharge side (as 3"-P-1011-B3; EQ-ep-5 resolution)
Design temperature	250 °C	as R-102
Code	ASME VIII Div. 1 (UG-27, UG-32)	typical
Material / S at design T	SA-516 Gr 70 (CS plate, killed, normalised) · 138 MPa	typical table (assumed)
Joint efficiency E / CA	1 / 3 mm	assumed
Shell: required / plate	14.9 / 16 mm	t = PR/(SE − 0.6P) + CA
Heads 2:1: required / plate	14.7 / 18 mm	t = PD/(2SE − 0.2P) + CA, +10 % forming
MAWP (corroded, hot)	42.7 bar g	from the plates chosen
Hydrotest (UG-99(b))	52.0 bar g at the top	1.3 × design × S(test)/S(design)
Candles	48 × Ø60 × 1500 mm sintered 316L, 13.6 m²	assumed
Weight (estimate)	each: empty 1.6 t	each; steel × 1.6 (cover flange, candles) (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
N1	Slurry inlet from P-102A/B	3"	300 RF	3"-P-1011-B3	3"-SL-2002	40 bar g / 250 °C
N2	Filtrate outlet to LV-102 / R-103	3"	300 RF	3"-P-1012-B1	3"-P-2003	40 bar g / 250 °C
N3	Cake discharge / back-flush (bottom)	4"	300 RF	—	—	40 bar g / 250 °C
N4	N_2 blow-back / vent	2"	300 RF	—	—	40 bar g / 250 °C
K1	PDT-102 taps (2)	1"	300 RF	—	—	40 bar g / 250 °C
K2	TSV-102F	1"	300 RF	—	—	40 bar g / 250 °C

Note: Design 40 bar g = the P-102 discharge side; TSV-102F to be set at 40 bar g (EQ-em-11).

CHX-V1-ME-DS-P-100 · Benzene feed pumps P-100A/B

Item	Value
Tag / quantity	P-100A/B (2 × 100 %)
Service	TK-100 → FIC-101/XV-101 → R-101
Type	Centrifugal, multistage between-bearings (API 610 BB3/BB4 type, typical)
Size / capacity	rated 48.7 m³/h at 374 m · ΔP 31.4 bar; absorbed ≈ 71 kW (η 0.60 assumed)
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-06 (CHX-V1-PC-PKG-001)
Driver	P-100A/B: 110 kW

Process data

Quantity	Value	Source
Liquid	benzene	
Rated flow	48.7 m³/h (41.8 t/h)	120 % case; PI-101 + 0.5 bar lines + 1.5 bar FV-101 (assumed) − suction 1.0 bar g
Density	858 kg/m³	HMB
Suction / discharge (design)	1.00 / 30.00 bar g	HMB
Differential / head	31.44 bar / 373.6 m	rated
Temperature	40.0 °C	HMB
Absorbed power estimate	70.9 kW (η 0.6 assumed); model econ() 90 kW	EQ-ep-10
Motor	110 kW, 0.4 kV	CHX-V1-EE-LL-001

Mechanical data

Quantity	Value	Basis
Design pressure	40.0 bar g	casing = discharge line 3"-P-1002-B1 (40 bar g)
Design temperature	70 °C	as the line
Type	Centrifugal, multistage between-bearings (API 610 BB3/BB4 type, typical)	
Materials / seals	Casing CS (S-6 class, API 610 typical); dual mechanical seals (plan 53B, assumed)	
Weight (estimate)	3.5 t each	pump + motor + baseplate, each (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
S	Suction	6"	300 RF	6"-P-1001-B1	6"-P-1001	40 bar g / 70 °C
D	Discharge (to the common header)	3"	300 RF	3"-P-1002-B1	4"-P-1002 ≠	40 bar g / 70 °C

CHX-V1-ME-DS-P-101 · Pump-around pumps P-101A/B

Item	Value
Tag / quantity	P-101A/B (2 × 100 %)
Service	R-101 → FV-104 → E-101 / by-pass → R-101 (always K.Fpa)
Type	Centrifugal, overhung centreline-mounted (API 610 OH2 type, typical), slurry service
Size / capacity	rated 998 m³/h at 33.6 m · ΔP 2.0 bar; absorbed ≈ 79 kW (η 0.70 assumed); NPSHa ≈ 3.9 m (Process engineering)
P&ID / package	CHX-V1-PID-001 sheet 1 · PK-06 (CHX-V1-PC-PKG-001)
Driver	P-101A/B: 200 kW

Process data

Quantity	Value	Source
Liquid	cyclohexane + Raney-Ni slurry	
Rated flow	997.8 m³/h (606.0 t/h)	P-101 rated flow = K.Fpa; ΔP 2.0 bar (Process tab streamRows)
Density	607 kg/m³	HMB
Suction / discharge (design)	30.00 / 32.00 bar g	HMB
Differential / head	2.00 bar / 33.6 m	rated
Temperature	200.0 °C	HMB
Absorbed power estimate	79.2 kW (η 0.7 assumed); model econ() 160 kW	EQ-ep-10
Motor	200 kW, 0.4 kV	CHX-V1-EE-LL-001

Mechanical data

Quantity	Value	Basis
Design pressure	40.0 bar g	as 16"-P-1004-B3
Design temperature	250 °C	SL_ASSUMED.Tdes1
Type	Centrifugal, overhung centreline-mounted (API 610 OH2 type, typical), slurry service	
Materials / seals	Casing CS + hard-faced wear parts (slurry); dual pressurised seals (plan 53B/54, typical for hot flashing slurry)	
Weight (estimate)	5.0 t each	pump + motor + baseplate, each (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
S	Suction	18"	300 RF	18"-P-1003-B3	18"-SL-1010	40 bar g / 250 °C
D	Discharge	16"	300 RF	16"-P-1004-B3	16"-SL-1011	40 bar g / 250 °C

CHX-V1-ME-DS-P-102 · R-102 transfer pumps P-102A/B

Item	Value
Tag / quantity	P-102A/B (2 × 100 %)
Service	R-102 → F-102A/B → LV-102 → R-103
Type	Centrifugal, overhung (API 610 OH2 type, typical), slurry service
Size / capacity	rated 39.5 m³/h at 48 m · ΔP 3.0 bar (K.pumpDP); absorbed ≈ 6.6 kW (η 0.50 assumed)
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-06 (CHX-V1-PC-PKG-001)
Driver	P-102A/B: 37 kW

Process data

Quantity	Value	Source
Liquid	cyclohexane + Raney-Ni slurry	
Rated flow	39.5 m³/h (25.3 t/h)	line 4"-P-1010-B3 rated flow; ΔP = model K.pumpDP
Density	641 kg/m³	HMB
Suction / discharge (design)	29.60 / 32.60 bar g	HMB
Differential / head	3.00 bar / 47.7 m	rated
Temperature	160.0 °C	HMB
Absorbed power estimate	6.6 kW (η 0.5 assumed); model econ() 30 kW	EQ-ep-10
Motor	37 kW, 0.4 kV	CHX-V1-EE-LL-001

Mechanical data

Quantity	Value	Basis
Design pressure	40.0 bar g	as 3"-P-1011-B3
Design temperature	250 °C	SL_ASSUMED.Tdes1
Type	Centrifugal, overhung (API 610 OH2 type, typical), slurry service	
Materials / seals	Casing CS + hard-faced wear parts; dual seals (typical)	
Weight (estimate)	1.2 t each	each (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
S	Suction	4"	300 RF	4"-P-1010-B3	6"-SL-2001 ≠	40 bar g / 250 °C
D	Discharge	3"	300 RF	3"-P-1011-B3	3"-SL-2002	40 bar g / 250 °C

CHX-V1-ME-DS-K-101 · Recycle hydrogen compressor

Item	Value
Tag / quantity	K-101 (1)
Service	V-104 gas → FIC-103/FV-103 → R-101 sparger
Type	Centrifugal, 1–2 impellers, dry gas seals (typical for low-ratio recycle H_2; type to be confirmed, EQ-ee-1)
Size / capacity	46,000 Nm^3/h (120 %: 55,200); ratio 1.100 · head 81.6 kJ/kg; absorbed ≈ 194 kW (η 0.75, k 1.4 assumed) vs model econ() 1,200 kW
P&ID / package	CHX-V1-PID-001 sheet 2 · PK-07 (CHX-V1-PC-PKG-001)
Driver	K-101: 1600 kW, 6.6 kV

Process data

Quantity	Value	Source
Gas	recycle H_2 92.1 % (rest CH_4), MW 3.13 (dry; saturated MW differs, EQ-ep-8)	HMB 18
Flow	46,000 Nm^3/h = 6.42 t/h (120 %: 55,200 Nm^3/h)	FIC-103
Suction	29.00 bar g · 44.9 °C	HMB
Discharge	32.00 bar g · 60 °C (model gas temperature)	HMB
Ratio / head	1.100 / 81.6 kJ/kg (k 1.4)	assumed k
Absorbed power estimate	194 kW design, 179 kW at 120 % (η 0.75); model econ() 1,200 kW	EQ-ee-1, EQ-ep-10
Motor	1600 kW, 6.6 kV	CHX-V1-EE-LL-001

Mechanical data

Quantity	Value	Basis
Design pressure	42.0 bar g	SL_ASSUMED.PdesK = PSV-112 set (deadhead from 35 bar g suction × (1 + K.kk))
Design temperature	90 °C	as 8"-H-2008-B2
Type	Centrifugal, 1–2 impellers, dry gas seals (typical for low-ratio recycle H_2; type to be confirmed, EQ-ee-1)	
Materials	Casing CS (killed); impellers 17-4PH (typical)	
Protection	PSV-112 on the discharge (42 bar g), NRV-112, anti-surge recycle (to be confirmed)	Safety layers
Weight (estimate)	35 t	package incl. motor, seal and lube system (assumed)

Nozzle schedule

Mark	Service	Size	Class	Line (LL-001)	EPC P&ID line	Design
S	Suction from V-104	8"	300 RF	8"-H-2007-B2	10"-G-2007 ≠	42 bar g / 90 °C
D	Discharge to FV-103	8"	300 RF	8"-H-2008-B2	8"-G-2008	42 bar g / 90 °C